TUBA1C-related Preimplantation Embryonic Arrest (OZEMA24)

Comprehensive Disease Characterization Report

Causal gene: *TUBA1C* (tubulin alpha 1c; HGNC:20768). This report concerns the Mendelian disorder in which **biallelic (homozygous) loss-of-function variants in the α-tubulin isotype gene *TUBA1C*** cause female infertility presenting as recurrent preimplantation embryo arrest. Identifiers: **MONDO:0979232**, **OMIM:621232 (OZEMA24 — "oocyte/zygote/embryo maturation arrest 24")**. The disease is distinct from, but mechanistically parallel to, the better-known β-tubulin (*TUBB8*) oocyte/embryo-arrest disorders. Note also that a second α-tubulin gene, *TUBA4A*, was reported in the same study for RPEA, but *TUBA4A* is separately and predominantly associated with neurodegenerative/myopathic phenotypes (ALS/FTD, SPAX11, CMYO26) — do not conflate the two.

> **Evidence base.** The disease entity rests almost entirely on a single primary human-genetics paper: **Hu H, Wan X, Zhang H, et al. "Biallelic variants in α-tubulin isotypes cause female infertility characterised as recurrent preimplantation embryo arrest." *J Med Genet* 2024 Oct 23;61(11):1045–1052 (** P 39209701 **).**
> All TUBA1C-specific claims derive from this report; broader claims are supported by the tubulin/RPEA literature and are flagged as analogy/inference where appropriate.

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1. Disease Information

- **Overview.** A rare Mendelian cause of female infertility in which fertilized eggs or early cleavage-stage embryos repeatedly stop developing before blastocyst formation/implantation, yielding no transferable embryos during IVF/ICSI. It belongs to the family of **recurrent preimplantation embryo arrest (RPEA / PREA / PREMBA)** disorders caused by defects in genes governing the oocyte meiotic and early-embryonic mitotic spindle.

- **Key identifiers.**
- OMIM: **621232** (phenotype, OZEMA24)
- MONDO: **MONDO:0979232** — label "oocyte/zygote/embryo maturation arrest 24" (confirmed via EBI OLS4)
- MedGen: **UID 1876453**; UMLS: **C6012735** (MONDO equivalentTo cross-references)
- Gene: *TUBA1C* — **HGNC:20768**, **NCBI Gene 84790**, **Ensembl ENSG00000167553**, UniProt **Q9BQE3**
- Orphanet / ICD-10 / ICD-11 / MeSH: No dedicated disease-level code identified for this ultra-new entity; it falls under female infertility (ICD-10 N97.x; ICD-11 GA31) and "female infertility of other origin." MeSH: closest terms "Infertility, Female" and "Embryo Loss."
- **Synonyms / alternative names.** OZEMA24; TUBA1C-related recurrent preimplantation embryo arrest; α-tubulin (TUBA1C)-related oocyte/embryo maturation arrest.
- **Information source.** Derived from **individual patients** (whole-exome sequencing of an RPEA cohort with functional validation), not from aggregated disease-level registries. The knowledge base is a small number of index cases.

2. Etiology

- **Primary cause — genetic.** Homozygous loss-of-function variants in *TUBA1C*. Two variants are reported (Hu et al. 2024): **p.Gln358Ter** (nonsense/stop-gain) and **p.Asp444Metfs*42** (frameshift). Both are predicted loss-of-function and both were homozygous. Verbatim: *"Four homozygous variants were identified in the PREA cohort: two of TUBA1C (p.Gln358Ter and p.Asp444Metfs*42) ... These variants cause varying degrees of spindle assembly defects"* (P 39209701).
- **Genetic risk factors.** The causal variants themselves are the risk determinant. **Consanguinity / homozygosity** is implicit (homozygous variants in a recessive disorder). Carrier (heterozygous) parents are unaffected. No susceptibility loci or modifier genes have been defined for this entity.
- **Environmental / lifestyle / infectious factors. None known or implicated.** This is a monogenic cytoskeletal disorder; there is no evidence for toxic, infectious, or lifestyle contributions. (Not applicable.)

- **Protective factors.** None defined. By mechanism, presence of at least one functional *TUBA1C* allele (heterozygous carrier state) is "protective" (asymptomatic), consistent with recessive inheritance.
- **Gene–environment interactions.** None described. (Not applicable with current evidence.)

3. Phenotypes

The phenotype is **reproductive**, detected only in the context of attempted conception/ART. There are no somatic/systemic manifestations reported.

Phenotype	Type	Onset	Severity	Frequency	Suggested HPO
Female infertility	Clinical sign	Reproductive age	Severe	100% of affected	**HP:0008222** (Female infertility) / HP:0000133
Recurrent preimplantation embryo arrest (no blastocyst)	Lab/clinical (ART)	At IVF/ICSI	Severe	100%	HP:0030033 (Abnormal fertility) — no precise HPO; closest HP:0100678? N/A
Repeated IVF/ICSI failure (no transferable embryos)	Clinical observation	At ART	Severe	100%	—
Meiotic spindle / early-cleavage spindle assembly defect	Lab/cellular	Oocyte–zygote	Variable ("varying degrees")	—	—

- **Characteristics.** Onset is effectively **congenital at the gametic level** but clinically manifests only in **adulthood at reproductive age** when ART is attempted. Course is **stable/non-progressive** (a fixed reproductive block, not a degenerative disease). Severity is **severe** for the reproductive endpoint (recurrent failure).

- **Quality-of-life impact.** Substantial psychological and emotional burden typical of infertility and recurrent ART failure (anxiety, depression, relationship/financial strain), but **no physical disability or systemic illness**. No disease-specific QoL instrument data exist for TUBA1C; general infertility QoL tools (FertiQoL, SF-36) would apply.

4. Genetic / Molecular Information

- **Causal gene.** *TUBA1C* — tubulin alpha 1c (12q13.12). One of the human α-tubulin isotypes forming the obligate α/β-tubulin heterodimer, the building block of microtubules.
- **Pathogenic variants (reference transcript NM_032704.5; both ClinVar "Pathogenic" for OZEMA24):**
- **p.Gln358Ter** (ClinVar "Q358*", Variation ID 3901262) — nonsense (stop-gain) → predicted loss of function; ClinVar Pathogenic* (ACMG PVS1 + recessive segregation).
- **c.1330del → p.Asp444Metfs*42** (ClinVar "TUBA1C, 1-BP DEL, NT1330", Variation ID 3901263) — single-base-deletion frameshift → truncation/loss of function; **ClinVar Pathogenic**.
- Both are annotated in ClinVar to the condition **"Oocyte/zygote/embryo maturation arrest 24" (OZEMA24)**. The remainder of the TUBA1C ClinVar entry set is dominated by variants of uncertain significance (VUS), underscoring the novelty of these two LOF alleles.
- Both **germline**, **homozygous (biallelic)**, **maternal-effect/embryonic**. No somatic involvement.
- **Functional consequence. Loss of function** (truncation/NMD-susceptible transcripts, loss of a functional α-tubulin isotype) leading to impaired spindle assembly. (Contrast: *TUBB8* disease variants are typically **dominant-negative missense** that poison heterodimer assembly.)
- **Allele frequency.** The specific LOF variants are ultra-rare/absent or near-absent in population databases (gnomAD) consistent with a severe recessive phenotype.
- **Gene constraint (gnomAD, this analysis).** *TUBA1C* pLI = **0.055**, observed/expected LoF (oe_lof) = **0.458** (90% CI 0.31–0.70), missense Z = **4.55** (strongly missense-constrained), lof_z = 2.72. The tolerance of heterozygous LoF explains the **autosomal recessive** mode (carriers unaffected); strong missense constraint reflects structural sensitivity of the folded tubulin.
- **Modifier genes / epigenetics / chromosomal abnormalities.** None established for TUBA1C. (Of note, downstream single-cell RNA-seq of arrested embryos in the companion *TUBA4A* cases showed altered spindle organisation, chromosome segregation and **mRNA decay** programs — relevant to mechanism, not a primary chromosomal lesion.)

5. Environmental Information

Not applicable. No environmental, lifestyle, toxic, or infectious factors contribute to this monogenic cytoskeletal disorder.

6. Mechanism / Pathophysiology

Causal chain (initiating lesion → clinical manifestation)

1. **Biallelic LOF variant in *TUBA1C*** (p.Gln358Ter nonsense; p.Asp444Metfs*42 frameshift) → **leads to** loss/truncation of functional TUBA1C α-tubulin protein (likely via nonsense-mediated decay and/or non-functional truncated polypeptide). *[demonstrated: variants identified; LOF inferred from variant class]*
2. Reduced functional α-tubulin pool → **results in** defective assembly of the **α/β-tubulin heterodimer** and impaired microtubule polymerization in the oocyte/early embryo. *[inferred from tubulin biology; TUBB8 paradigm]*
3. Impaired microtubule assembly → **leads to meiotic and early-mitotic spindle assembly defects** ("varying degrees of spindle assembly defects,"

   P 39209701

). *[demonstrated in vitro]*
4. Spindle defects → **cause chromosome mis-segregation / mitotic failure** and dysregulated maternal **mRNA decay / zygotic transition** programs in cleavage-stage embryos. *[demonstrated by single-cell RNA-seq in the companion α-tubulin (TUBA4A) cases; inferred for TUBA1C]*
5. Failed cell division → **results in arrest of preimplantation embryo development** before the blastocyst stage (no viable/transferable embryo). *[clinical observation]*
6. → **Clinical manifestation: recurrent IVF/ICSI failure and female infertility.**

Branch point: the lesion can manifest as (a) meiotic spindle failure at the oocyte stage and/or (b) first-cleavage mitotic spindle failure in the zygote/early embryo; the reported TUBA1C phenotype centers on **post-fertilization embryonic arrest** rather than pre-fertilization oocyte-maturation arrest, distinguishing it along the spectrum seen with TUBB8.

Checklist coverage

- **Molecular pathways / processes.** Microtubule cytoskeleton assembly; mitotic/meiotic spindle organization; cell-cycle progression (M-phase); chromosome segregation; maternal mRNA clearance / maternal-to-zygotic transition. **GO terms:** GO:0007017 (microtubule-based process), GO:0000226 (microtubule cytoskeleton organization), GO:0007052 (mitotic spindle organization), GO:0007051 (spindle organization), GO:0051301 (cell division), GO:0007059 (chromosome segregation), GO:0045132 (meiotic chromosome segregation).
- **Protein dysfunction.** Loss of function of an α-tubulin isotype; impaired heterodimer formation and microtubule nucleation/dynamics. (CHEBI: GTP CHEBI:15996 — tubulin is a GTP-binding protein; paclitaxel-class CHEBI not therapeutically relevant here.) **Structural detail (UniProt Q9BQE3, 449 aa):** both variants converge on the **C-terminal acidic "E-hook" tail** (disordered region residues 429–449; sequence …KDYEEVGADSADGEDEGEEY). This tail carries regulatory PTM sites (phospho-Tyr432, phospho-Ser439) and the terminal **Tyr449** that undergoes the detyrosination/tyrosination cycle. **p.Gln358Ter** deletes ~92 C-terminal residues (~20% of the protein, removing the entire E-hook and part of the folded core); **p.Asp444Metfs*42** spares the folded core but replaces the extreme C-terminal tail with 42 aberrant residues. The E-hook mediates binding of microtubule-associated proteins (MAPs), molecular motors, and +TIPs — so its loss impairs microtubule–MAP/motor interactions and spindle regulation in addition to any NMD-mediated null effect.
- **Cellular processes.** Cell-cycle/M-phase dysregulation, failed cytokinesis, developmental arrest (a form of division failure rather than classic apoptosis, though embryonic fragmentation/apoptosis can ensue).
- **Metabolic / immune / infectious involvement.** None. (Not applicable.)
- **Molecular profiling.** Single-cell/single-embryo RNA-seq of arrested embryos (companion α-tubulin cases) highlighted spindle organization, chromosome segregation and mRNA-decay transcriptomic disruption.
- **Expression note (mechanistic nuance).** Human Protein Atlas: *TUBA1C* RNA is **"detected in all" tissues** and localizes to **microtubules** — i.e., TUBA1C is broadly/ubiquitously expressed, unlike oocyte-specific *TUBB8*. The oocyte/early-embryo-restricted phenotype despite ubiquitous expression implies the **uniquely error-intolerant meiotic/first-mitotic spindle** is the vulnerable window, where TUBA1C contributes substantially and other α-tubulin isotypes incompletely compensate.

Cell types (CL): oocyte **CL:0000023**, zygote/fertilized egg **CL:0000365**, embryonic cell/blastomere **CL:0000353**.

7. Anatomical Structures Affected

- **Organ level.** Primary: the **oocyte and preimplantation embryo** (female reproductive system). No secondary organ involvement. Body system: **reproductive system** only.
- **Tissue / cell level.** Germ cells — oocytes (**CL:0000023**), and the cleavage-stage embryo/blastomeres (**CL:0000353**); no somatic tissue pathology.
- **Subcellular level. Microtubule cytoskeleton / meiotic & mitotic spindle** (GO cellular component GO:0005874 microtubule; GO:0005819 spindle; GO:0072686 mitotic spindle). Localization confirmed to microtubules (HPA).
- **Localization (UBERON).** Ovary **UBERON:0000992**; oocyte-containing follicle; embryo **UBERON:0000922**. **Bilateral** (both ovaries / all oocytes affected); the defect is intrinsic to every gamete/embryo (not focal/lateralized).

8. Temporal Development

- **Onset.** The molecular lesion is **congenital/germline**; the clinical phenotype has an **adult (reproductive-age) presentation**, discovered only when pregnancy/ART is attempted. Onset pattern of the embryonic event is **acute per cycle** (arrest occurs within days of fertilization, typically cleavage stage, before blastocyst).
- **Progression. Non-progressive and stable** — a fixed reproductive block; not a staged or degenerative disease. The pattern is **recurrent/reproducible across IVF cycles** (hence "recurrent" embryo arrest).
- **Critical period / window of intervention.** The vulnerable window is **oocyte meiosis through the first few embryonic cleavage divisions**. There is no therapeutic window to rescue the patient's own oocytes; the only effective "intervention point" is to bypass the gametes (oocyte donation).

9. Inheritance and Population

- **Inheritance pattern. Autosomal recessive** (biallelic/homozygous LOF; unaffected heterozygous carriers). Consistent with gnomAD LoF tolerance (pLI 0.055).
- **Penetrance / expressivity.** Apparently **high penetrance** for the reproductive phenotype in homozygotes; **variable expressivity** at the cellular level ("varying degrees of spindle assembly defects"). No anticipation (not a repeat-expansion disorder).

- **Carrier frequency / founder effects / consanguinity.** Not formally estimated; homozygous presentation suggests **consanguinity or population homozygosity**. Cohort of Chinese ancestry (study from Central South University / Shandong University). No founder haplotype defined.
- **Epidemiology. Ultra-rare.** RPEA overall is uncommon; TUBA1C accounts for only a tiny minority of RPEA cases (two variants reported to date). No prevalence/incidence figures exist. **Sex:** manifests in **females** (maternal/embryonic effect); males homozygous for LOF have no reported reproductive phenotype attributable to TUBA1C in this study. **Geographic data:** index cases East Asian; too few to define distribution.

10. Diagnostics

- **Primary diagnostic route — genetic. Whole-exome sequencing (WES)** of the infertile woman (the discovery method), confirmed by **Sanger sequencing**; segregation in parents (carrier status). **WGS** or **targeted RPEA/oocyte-maturation-arrest gene panels** (which should now include *TUBA1C* alongside *TUBB8, TUBA4A, PATL2, WEE2, TLE6, NLRP5, PADI6, KPNA7, GMNN, HFM1, MEI1, KIF11, CKAP5*) are the practical diagnostic tools. Single-gene testing is appropriate once a panel/WES flags *TUBA1C*.
- **Clinical/ART findings (phenotypic diagnosis).** Documentation at IVF/ICSI of **repeated embryonic arrest with no blastocyst/transferable embryo** despite normal oocyte retrieval; this ART phenotype prompts genetic testing. Time-lapse embryo morphokinetics can document the arrest.
- **Supportive research assays.** In vitro heterodimer-assembly/microtubule assays, immunofluorescence of spindle architecture, and **single-embryo/single-cell RNA-seq** (research, not clinical).
- **Not applicable:** CMA/karyotype/FISH (no structural chromosomal lesion), mtDNA testing, repeat-expansion testing, routine biochemistry/imaging — all normal/uninformative.
- **Differential diagnosis.** Other maternal-effect/spindle genes causing oocyte-maturation arrest, fertilization failure, or embryo arrest — especially ***TUBB8*** (dominant, oocyte-specific), ***TUBA4A***, ***KIF11***, ***CKAP5*** (spindle), and zygotic-genome-activation/cell-cycle genes (***KPNA7***, ***GMNN***, ***HFM1***, ***MEI1***, ***PADI6***, ***NLRP5***, ***TLE6***, ***WEE2***, ***PATL2***). Distinguished by gene identity, inheritance mode, and the precise developmental stage of arrest.
- **Screening.** No population newborn/carrier screening. **Cascade carrier testing** of relatives and **preconception carrier screening** in consanguineous couples are reasonable once a familial variant is known.

11. Outcome / Prognosis

- **Survival/mortality: Not applicable** — no effect on patient health or lifespan; the disease affects reproduction only.
- **Reproductive prognosis with own oocytes: Poor** — recurrent IVF/ICSI failure; by analogy to TUBB8/KIF11, **no viable pregnancies** have been documented with the patient's own gametes.
- **Prognosis with intervention: Good via oocyte donation** (bypasses the defective maternal gametes).
- **Morbidity:** Limited to infertility and its psychosocial impact; no physical disability.
- **Prognostic factors:** Variant severity (complete LOF) and the stage of arrest; essentially uniformly severe for the autologous-gamete endpoint.

12. Treatment

- **No pharmacological or gene/cell/RNA therapy exists**; no targeted drug. Microtubule-targeting drugs are not therapeutic here.
- **Standard of care: Genetic counseling + third-party reproduction (donor-oocyte IVF).** Supporting evidence (spindle-gene RPEA generally): Voros et al. 2025 (P 40650169): *"Donor egg IVF was the sole effective treatment; however, no viable pregnancies were documented in patients with pathogenic mutations of TUBB8 or KIF11."* Applied to TUBA1C by mechanistic analogy (no TUBA1C-specific outcome data yet).
- **Supportive care:** psychological support for infertility.
- **Experimental:** None specific. (NCIT suggestions: **In Vitro Fertilization** NCIT:C15387; **Oocyte Donation** — donor-gamete ART; **Genetic Counseling** NCIT:C15499.)
- **Pharmacogenomics / immunotherapy / surgery:** Not applicable.

13. Prevention

- **Primary prevention: Preconception / carrier genetic screening**, especially in consanguineous couples; **genetic counseling** with reproductive-risk assessment (25% recurrence for two carriers).

- **Reproductive options to avoid affected outcomes: Oocyte donation** (most effective); **PGT** is generally **not useful** for a maternal-effect/embryonic cytoskeletal defect because the patient's own embryos arrest before testing/transfer — so prevention centers on gamete donation rather than embryo selection.
- **Secondary/tertiary prevention, immunization, public-health/environmental measures:** Not applicable (monogenic, non-infectious, non-environmental).

14. Other Species / Natural Disease

- **Taxonomy / orthologs.** Human *TUBA1C* (**NCBI Gene 84790**). Mouse ortholog ***Tuba1c*** (and the closely related *Tuba1a/Tuba1b* cluster). α-tubulin is **deeply evolutionarily conserved** across eukaryotes; the α/β-tubulin spindle mechanism is universal. **HomoloGene/Alliance** confirm orthology.
- **Natural disease in other species: No naturally occurring TUBA1C-related embryo-arrest disease is catalogued** (e.g., not in OMIA) for companion animals/wildlife. (Not applicable with current data.)
- **Comparative biology:** The requirement of α-tubulin for meiotic/mitotic spindles is conserved, making the mechanism broadly generalizable, but the specific oocyte/embryo-restricted human phenotype reflects isotype usage and redundancy particular to primate/human early development.
- **Transmission / zoonosis:** Not applicable (genetic disease).

15. Model Organisms

- **No dedicated published *Tuba1c*-knockout model of embryo arrest** was identified. Functional validation to date is **in vitro / in cultured cells** and via **single-cell RNA-seq of human arrested embryos** (P 39209701).
- **Constitutive mouse *Tuba1c* knockout does NOT recapitulate the human disease.** IMPC single-gene knockout data report only **"increased circulating alkaline phosphatase level"** (homozygote), with **viable homozygotes and no infertility/embryo-arrest phenotype**. This is a major model limitation, most plausibly reflecting **differential α-tubulin isotype redundancy** between mouse and human oocytes/early embryos (paralleling the primate-

specificity of *TUBB8*). Oocyte-specific/humanized models or human iPSC-derived gamete/embryo systems would be required to recapitulate the phenotype.

- **Relevant comparative models (paradigm genes):** *TUBB8* mutations have been studied in **cultured HeLa cells, yeast (microtubule dynamics), and mouse oocytes** (P 26789871) (P 27273344) — these recapitulate spindle-assembly disruption and establish the tubulin-spindle mechanism. Mouse *Tuba1c* gene-targeting (IMPC/MGI) could serve as a future model; conditional/oocyte-specific knockouts would best recapitulate the human reproductive phenotype.
- **Model limitations:** *TUBB8* is primate-specific (no true mouse ortholog), so mouse models imperfectly capture isotype-specific human oocyte biology; by extension, modeling the exact human TUBA1C contribution requires humanized or human-embryo systems (iPSC-derived gametes/organoids are emerging but not yet established for this gene).
- **Resources:** MGI (*Tuba1c*), IMPC/IMSR, Alliance of Genome Resources.

Supported vs. Refuted Hypotheses

Hypothesis	Status	Basis
Biallelic *TUBA1C* LOF causes recurrent preimplantation embryo arrest	**Supported**	P 39209701 : two homozygous LOF variants, spindle defects
Mechanism is α/β-tubulin spindle disruption (shared with TUBB8)	**Supported**	Functional assays; tubulin-spindle paradigm (26789871, 40650169)
Inheritance is autosomal recessive	**Supported**	Homozygous variants; gnomAD LoF tolerance (pLI 0.055)
Donor-oocyte IVF is the only effective management	**Supported (by analogy)**	P 40650169 for TUBB8/KIF11; inferred for TUBA1C
Environmental/lifestyle factors contribute	**Refuted / N.A.**	Monogenic cytoskeletal disorder; no evidence
Disease causes systemic/somatic illness	**Refuted**	Phenotype restricted to reproduction despite ubiquitous expression

Limitations and Future Directions

- The entire TUBA1C-specific evidence base is **one paper with two families/variants**; penetrance, expressivity, carrier frequency, and population distribution are essentially unknown.
- No animal model, no clinical trial, and no TUBA1C-specific treatment-outcome data exist.
- The precise explanation for an **oocyte/embryo-restricted phenotype from a ubiquitously expressed gene** (isotype redundancy, dosage, developmental spindle vulnerability) remains to be dissected.
- **Future work:** additional patient ascertainment to refine the variant/phenotype spectrum; oocyte-specific *Tuba1c* mouse or human iPSC-gamete/organoid models; integration into

clinical RPEA gene panels; and formal definition of the developmental stage(s) of arrest by time-lapse and single-cell multi-omics.

Key Citations

- PMID 39209701 — Hu et al., *J Med Genet* 2024;61(11):1045–1052. Primary disease-defining paper (TUBA1C p.Gln358Ter, p.Asp444Metfs*42; spindle defects; RPEA).
- PMID 40650169 — Voros et al., *Int J Mol Sci* 2025;26:6390. Systematic review (TUBB8/KIF11/CKAP5); donor-egg IVF as sole effective treatment.
- PMID 26789871 — Feng et al., *N Engl J Med* 2016. TUBB8 and human oocyte meiotic arrest (mechanistic paradigm).
- PMID 27273344 — Feng et al., 2016. TUBB8 phenotypic spectrum (oocyte/early-embryo).
- PMID 40455380 / **36647821** / **41423819** — GMNN / KPNA7 / HFM1 RPEA genes (differential diagnosis context).
- Gene/constraint data: Ensembl ENSG00000167553; gnomAD (pLI 0.055, oe_lof 0.458, mis_z 4.55); Human Protein Atlas (ubiquitous, microtubule localization).